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A gravitational-wave measurement of the Hubble constant following the second observing run of Advanced LIGO and Virgo

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ABSTRACT

Measurement H0 = 69 (+16, -8) km s^-1 Mpc^-1 (68.3% HPDI, flat-in-log prior) from GWTC O1+O2 using standard sirens: GW170817 with EM counterpart + binary black holes (BBH) with galaxy catalogs. Improvement 4% over GW170817 alone (69 +17 -8). Significant contribution from GW170814 well-localized and covered by DES. Uses Planck 2015 cosmology Omega_m=0.308, Omega_Lambda=0.692 for host weighting. Method: luminosity distance from GW waveform + redshift from catalog (GLADE) or counterpart.

What was actually measured (numbers):

H0 69 km/s/Mpc, interval [20,140] prior uniform, GW170817 bright siren + 42 BBH dark sirens, factor 1.04 improvement

Devices / Protocol:

Advanced LIGO Hanford/Livingston, Virgo, GLADE galaxy catalog, Dark Energy Survey (DES)

Replication across teams / Independence:

Independent from Planck and SH0ES; standard-siren route orthogonal to CMB and distance ladder

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